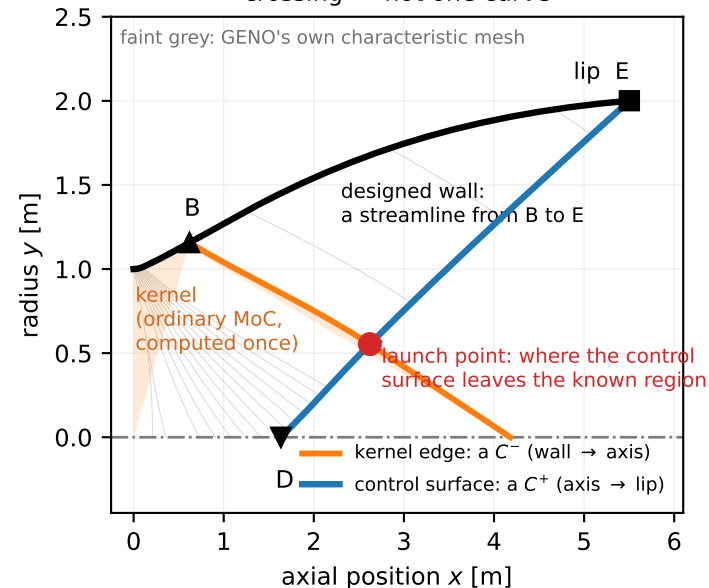
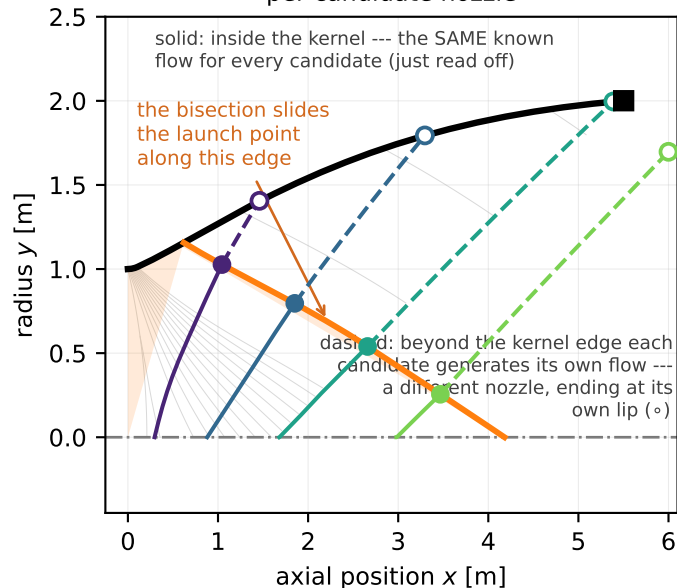


(a) the construction: two characteristic families, crossing --- not one curve



(b) the one-parameter family: one launch point per candidate nozzle



(c) why this one curve can be traced without the flow around it



ORDINARY characteristic

unknowns to advance: dq , $d\theta$
 relations available: compatibility only
 $\Rightarrow$ 2 unknowns, 1 equation --- does not close
so it cannot be walked alone: it needs a partner from the other family --- it needs the surrounding flow to already be known

RAO's CONTROL SURFACE

same unknowns: dq , $d\theta$
 relations available: compatibility
 + the optimality invariant
 $\Rightarrow$ 2 unknowns, 2 equations --- CLOSED

*it generates itself from one starting state:
 that is what makes the whole method possible*